ADDIS ABABA UNIVERSITY

COLLEGE OF TECHNOLOGY AND BUILT ENVIRONMENT

SCHOOL OF INFORMATION TECHNOLOGY AND ENGINEERING

# Campus Event Management System

Project Proposal

Group 5

**PREPARED BY: -**

1. **Anatoli Chala UGR/4369/16**
2. **Christian Elias UGR/8399/16**
3. **Menal Abdulkadir UGR/7907/16**
4. **Natnael henok UGR/9671/16**
5. **Ruth Tewodros UGR/7383/16**
6. **Selam Tewodros UGR/2331/16**
7. **Yordanos Bisrat UGR/3362/16**

**ADVISOR**: Nuniyat Kifle

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# ACRONYMS

1. CEMS-Campus event management system
2. IT-Information Technology
3. UI-User Interface
4. DB-Database
5. SMS-Short Message Service
6. HTML-Hypertext Markup Language
7. CSS-Casacading Style Sheets
8. JS-JavaScript

# ABSTRACT

The Campus Event Management System(CEMS) is a web-based platform designed to simplify the planning, organization, and coordination of events within Addis Ababa University. Currently, most events are managed manually through Telegram groups, posters, or word of mouth, which leads to poor communication, low attendance, and lack of proper tracking. The proposed system will automate event management by allowing organizers to create and manage events, while students can browse, register, and receive updates in real time.

CEMS will feature a centralized event calendar, user registration, notifications, and feedback collection. The project aims to improve efficiency, accessibility, and engagement across the campus community by replacing the existing informal process with a structured, data-driven, and user-friendly system. This proposal presents the analysis, objectives, feasibility study, and implementation plan for developing the Campus Event Management System.

# 1. INTRODUCTION

## 1.1 Background

In Addis Ababa University, various events such as club meetings, workshops and competitions are held regularly. These activities help students to grow academically, professionally, socially and personally. Most events are handled now using class announcements, notice boards, posters or casual social media platforms like telegram groups and channels.

While these methods help spread information, they are uncoordinated and inefficient. Event details can get lost in group chats, schedules can overlap, students may fall behind on important updates. Organizers also lack the ability to monitor attendance or get feedback. These issues present challenges for organizing event and limit the involvement of students.

The absence of an all-around digital platform is disappointing for the students, the organizers, and the administrators. The students are not able to find events or receive reminders in time. The organizers cannot handle registrations and send updates in an effective way. Administers have no formal way to approve, verify, and monitor events.

There is a need for a Campus Event Management Platform that will address these issues by bringing all university events into one integrated platform. The platform will enable event creation, scheduling, registration, reminders and gathering feedback. By centralizing event information, the system reduces mistakes, improves coordination, fosters student engagement, and increases transparency in our campus.

The idea for this platform comes from the need to make university life simpler and more enjoyable for students. At Addis Ababa University, events are an important part of campus life, but managing them manually often causes mistakes, wastes time, and makes the whole process harder for both students and organizers.

Creating a web-based Campus Event Management Platform will make it easy for students to find accurate, up-to-date information about events, while helping organizers save time and stay organized. This system will not only make events more structured and coordinated but also make the student experience more interactive and engaging. In the end, it will help build a stronger, more connected, and active university community.

### 1.2 The Existing System

Currently, campus events are organized in a very informal and tiring manner. Most announcements happen through telegram groups, posters, or word of mouth. There isn't a dedicated platform where students can see all upcoming events or register to participate. When registration is required, it usually happens manually via messages or in-person sign-ups.

While telegram helps to communicate quickly, it is not well-suited for event management. Important messages can easily get lost in chat threads, and there's no system to track attendance, send reminders, or maintain an organized event calendar. This lack of structure also makes it difficult to gather feedback or evaluate event success.

### 1.3 Statement of the Problem

The current process is disorganized and inefficient. Without a centralized system, information is scattered, communication is inconsistent, and event participation is hard to track.

The main problems include:

- No single platform to manage and view all campus events.
- Difficulty reaching all students effectively using telegram alone.
- No proper system for event registration, attendance tracking, or follow-up.
- Limited student engagement because updates often go unnoticed.

As a result, students often miss out on events, attendance is low, and organizers struggle to assess their events' effectiveness.

| Element | Description |
|---|---|
| **Why telegram isn't enough** | Telegram groups and channels, while convenient for chatting, are not designed for structured event management. The main shortcomings include:<br>**1.Unstructured event information**<br>**2.No registration feature**<br>**3.No automated reminders/notifications**<br>**4.Hard to find past or upcoming events**<br>**5.No analytics or attendance tracking**<br>**6.No approval process for official events** |
| **Affects ...** | The current disorganized method of managing events affects several key groups within the university:<br>1. **Students**: Often miss important updates or event opportunities due to scattered announcements and lack of reminders.<br>2. **Organizers and Clubs**: Struggle to coordinate effectively, track attendance, and gather feedback without a proper management tool.<br>3. **University Administration**: Lacks visibility into the types, frequency, and participation levels of campus events, making planning and reporting difficult.<br>**4. Department Heads and Faculty:** Have no unified way to announce academic seminars, workshops, or departmental activities.<br>**5. IT Support Staff**: Spend extra time responding to repetitive event-related inquiries that could be automated through a centralized system. |
| **Implementing the system will bring multiple benefits which include a centralized platform for easy access,improved communication between organizers and students,simplified registration and accurate tracking of attendees,insights from data to improve future events, and higher student participation and stronger campus management.** | The proposed **web-based Campus Event Management System will** serve as a single platform for creating, promoting, registering, and tracking events. Organizers can easily publish events, and students can browse, register, and receive notifications.<br>**Key features include:**<br>• Role-based access for students, organizers, and administrators.<br>• Tools for event creation and approval.<br>• Online registration and attendance tracking.<br>• Central event calendar with notifications.<br>• Feedback collection and analytics module. |

This system will turn the current manual and fragmented approach into a modern, organized, and interactive platform for campus-wide event management.

## 1.4 Objective of the Project

### 1.4.1 General Objective

To develop an automated campus event management system that accommodates overall management. That includes planning, organization, communication, and evaluation of all campus events at ADDIS ABABA UNIVERSITY. Thereby improves efficiency, coordination, and accessibility for event organizers and the whole campus community.

### 1.4.2 Specific Objective

- To design and develop a web-based event management system that allows event organizers to create, schedule, and manage events online.
- To automate notification and communication through email or SMS(e.g, event reminders, announcements).
- To enable campus students or staff to easily register for events, and have a real-time update for details of any event, such as time, venue, and agenda.
- Ensuring that everyone has the opportunity to seize any potential benefits from events.
- Integrating an approval workflow that makes the authentication of events by campus authorities much easier.
- To maintain a database for storing event information, participant information, and reviews, to make it a reference for better events in the future.

## 1.5 Proposed System

The proposed system is a web-based Campus Event Management System (CEMS) that automates and simplifies all event-related activities within the university. It will allow administrators, event organizers, and students to interact through a single platform.

Organizers will be able to create and manage events, add details such as date, venue, and agenda, and monitor participant registration. Students can view upcoming events, register, receive notifications, and give feedback after attending. The administrator will oversee event approvals, manage users, and ensure the quality and authenticity of event listings.

The system will be developed using modern web technologies such as HTML, CSS, JavaScript, Node.js, Express.js, and MongoDB. It will be hosted either on the university network or a free cloud-based service. By implementing this system, Addis Ababa University will move from a manual, fragmented process to a centralized, automated, and efficient event management environment, improving participation, communication, and data accuracy across the campus.

## 1.6 Feasibility Study

We carried out a feasibility study to evaluate whether our project is practical and has a good chance of success. This study helps us reduce risks and make well-informed decisions. To assess the likelihood of success, we examined several key feasibility criteria.

### 1.6.1 Economic Feasibility

This project is low-cost because we are using free and open-source tools. The main ones are Node.js, Express.js, MongoDB, HTML, CSS, and JavaScript.We will also use Visual Studio Code, Figma, and GitHub, which are all free to use. We can test the system on our own laptops without needing to pay for a server. So, the project is affordable and practical for us to build.

**Developmental cost**

We don't need to buy any software or special equipment for this project. Every group member already has a laptop and internet access. All the tools and resources we plan to use are free and easily available. The main investment for this project is our time and effort to design, build, and test the system. By using free tools and existing resources, we can focus more on learning and improving the project without worrying about extra costs.

**Operational cost**

After we finish the project, we can host it on a free website hosting service or on the school's network. It will not cost much to keep it running. Students or school administrators who have some technical knowledge can manage and maintain it easily. This makes the project both low-cost and simple to operate. Even if small problems happen, they can be fixed quickly by the team or by someone familiar with basic web tools.

## 1.6 Technical Feasibility

When we were choosing our project title, we focused only on ideas that we believe are possible to complete with our current skills. We are also taking a Web Programming course, which will help us design and build the project more easily. We plan to complete the project in three months, using both the skills we already have and the new skills we will learn along the way. If we face any challenges while developing the software, our group will discuss them and work together to find solutions.

### 1.6 Schedule Feasibility

We have about three months until the project deadline, and we have a clear plan to finish on time. Our project is simple and fits well with our skills and available time, so it is very likely that we will complete it as planned. We also include our time management plan later in this proposal.

### 1.7 Scope

Our system will include:

- Admin can add, edit, or delete events.
- The organizer can manage event details and participants.
- Students can view and register for events.
- Data will be stored in MongoDB.
- The system will be tested after each stage.

**Out of Scope**

The system will **not** include:

- A mobile app version.
- Online payment or ticket system.
- Email or SMS reminders.
- Calendar or Google integration.

### 1.8 Methodology

For this project, we decided to use the Agile model for the development process. The Agile model has easy and clear steps to follow from the planning design to coding to testing it and reviewing.

**Phases of development**

The first thing we did was get information about the current campus event management system. We talked to students and organisers and saw how they used spreadsheets and posters. Based on this, we made a list of the problems and the features that our system should have. These include signing up for events, getting approval from an admin, and getting notifications. Once we knew what we needed, we began to plan how the system would look and feel. We made diagrams and planned the layout of each page to show how data moves between users and the database.

The three main modules we will make are Admin, Organiser, and Student. We made the user interface with Figma and the diagrams with Draw.io. As soon as the design is done, the implementation will begin.

We will use:

- **HTML, CSS, and JavaScript** for the frontend
- **Node.js with express** for the backend
- **MongoDB** for the database

Each team member will work on a specific part of the system. We will use **GitHub** to share code and manage changes. Testing will be done throughout the development process, not just at the end.We will use:

- **Unit Testing** for individual components
- **Integration Testing** for combining module
- **User Testing** by students and organizers to check usability

In summary, we will use the Agile model to build our Campus Event Management System. The project will go through a number of sprints, each of which will include planning, designing, coding, testing, and getting feedback. This method helps us stay on track, work together, and build a system that meets the needs of users in a useful way.

## 1.9 Project Management plan

### 1.9.1. Time Management plan

The development of the Campus Event Management System will be completed in several major phases throughout the semester. Each phase has been allocated specific durations to ensure organized progress and balanced workload among team members. The time schedule below summarizes the key project tasks and their expected completion dates.

| No. | Task | Start Date | End Date | Duration | Status |
|---|---|---|---|---|---|
| 1 | Brainstorming and Idea Selection | Oct 2, 2025 | Oct 8, 2025 | 6 days | ✅ Completed |
| 2 | Title Approval | Oct 8, 2025 | Oct 11, 2025 | 3 days | ✅ Completed |
| 3 | Proposal Writing & Submission | Oct 12, 2025 | Oct 18, 2025 | 6 days | ✅ Completed |
| 4 | Software Requirements Specification (SRS) | Oct 19, 2025 | Oct 29, 2025 | 10 days | ⏳ Pending |
| 5 | Presentation Preparation (Proposal Defense) | Nov 5, 2025 | Nov 15, 2025 | 10 days | ⏳ Pending |
| 6 | Software Design Specification (SDS) | Nov 16, 2025 | Nov 26, 2025 | 10 days | ⏳ Pending |
| 7 | System Implementation and Testing | Nov 27, 2025 | Dec 20, 2025 | 24 days | 🔧 Planned |
| 8 | Final Presentation and Documentation | Dec 22, 2025 | Jan 4, 2026 | 14 days | 🔧 Planned |

Gantt Chart

| Task | Oct 1–10 | Oct 11–20 | Oct 21–31 | Nov 1–10 | Nov 11–20 | Nov 21–30 | Dec 1–10 | Dec 11–20 | Dec 21–Jan 4 |
|---|---|---|---|---|---|---|---|---|---|
| Brainstorming & Idea Selection | ⬜⬜⬜⬜⬜⬜⬜ | | | | | | | | |
| Title Approval | | ⬜⬜⬜ | | | | | | | |
| Proposal Writing & Submission | | ⬜⬜⬜⬜⬜⬜ | | | | | | | |
| SRS Documentation | | | ⬜⬜⬜⬜⬜⬜⬜ | | | | | | |
| Presentation Preparation | | | | ⬜⬜⬜⬜⬜⬜⬜ | | | | | |
| SDS Documentation | | | | | ⬜⬜⬜⬜⬜⬜⬜ | | | | |
| System Implementation & Testing | | | | | | ⬜⬜⬜⬜⬜⬜⬜⬜⬜ | ⬜⬜⬜⬜⬜⬜⬜⬜⬜ | | |
| Final Presentation & Documentation | | | | | | | | ⬜⬜⬜⬜⬜⬜⬜ | ⬜⬜⬜⬜⬜⬜⬜ |

### 1.9.2. Quality Management Plan

The quality of the Campus Event Management System is essential to ensure usability, performance, and reliability for both students and event organizers. Several potential risks could affect software quality during the project lifecycle. These include limited testing time, inconsistent feedback from users, integration errors between the front-end and back-end modules, and data validation issues that might compromise security or performance.

To manage these risks, the team will follow a structured quality assurance plan integrated into every development phase. Each build will be reviewed, tested, and validated before proceeding to the next stage. The quality management strategy emphasizes preventive measures rather than corrective actions.

#### Quality Assurance Approach

- **Code Review:** Peer reviews will be conducted after each feature is implemented to maintain consistency and readability.
- **Testing Process:** A multi-level testing process (unit, integration, and system testing) will be followed to identify issues early.
- **Version Control:** GitHub will be used to manage and track all code changes, ensuring traceability and preventing data loss.
- **User Feedback:** Feedback will be collected during the beta phase from selected students and event organizers to identify usability concerns.
- **Documentation Review:** All project documentation, including design and test reports, will be periodically reviewed to ensure accuracy.

#### Testing Considerations

- **Functional Testing:** Verify that event creation, updates, notifications, and registration functions operate as intended.
- **Integration Testing:** Ensure proper communication between the database, Node.js backend, and front-end interface.
- **Performance Testing:** Measure the system’s response time under different user loads.
- **Security Testing:** Validate input fields, prevent SQL injections, and ensure secure user login sessions.
- **Usability Testing:** Assess ease of navigation, readability, and layout clarity using student testers.
- **Acceptance Testing:** Final testing will confirm that the system meets all specified user requirements before deployment.

### 1.9.3. Communication Management Plan

Effective communication is crucial to the success of this project. The team will maintain consistent, transparent, and timely communication among members, supervisors, and external stakeholders. Internal communication ensures coordination during development, while external communication keeps advisors and users informed of progress and key updates.

#### Internal Communication

**Principle**: Ensure that all team members have access to current information about project progress, issues, and updates.

**Project Meetings**: Held weekly (in person or via Zoom) to discuss current tasks, risks, and upcoming milestones.

**Data Sharing:** All documents, source code, and reports will be stored in shared Google Drive folders and GitHub repositories.

**Sub-Group Reporting**: Front-end, back-end, and documentation sub-teams will report progress every week to the project coordinator.

**Milestone Meetings**: Conducted before key phases (proposal, SRS, design, testing) to evaluate progress and plan the next steps.

**Communication Channel:** A Telegram group will be used for quick daily communication, clarifications, and reminders.

#### External Communication

**Principle:** Keep instructors and potential users informed about project milestones, risks, and feedback.

**Project Reports:** A brief written progress summary will be submitted to the instructor every two weeks.

**Advisor Feedback Meetings:** Held twice a month to present progress and get guidance.

**User Testing Communication**: During beta testing, feedback forms and observation sessions will be organized for students and organizers

| Type of Communication | Method / Tool | Frequency /Schedule | Information | Participants / Responsibles |
|---|---|---|---|---|
| **Internal Communication:** | | | | |
| Project Meetings | In-person / Zoom | Weekly and as needed | Status updates, issues, next tasks | All Members |
| Sharing of project data | GitHub, Google Drive | Continuous | Code, documents, reports | Project Mgr(s) Project Team Members |
| Milestone Meetings | In-person | Before milestones | Progress review, feedback | Project Mgr Sub-project Mgr |
| Final Project Meeting | In-person | End of project | Wrap-up, evaluation | Project Mgr Project Team, Instructor |
| **External Communication and Reporting:** | | | | |
| Project Report | Excel sheet | Monthly | Project status<br>- progress<br>- forecast<br>- risks | Project Manager Sub-Project Managers |
| SteCo Meetings | Teleconference | Monthly | Progress discussion, strategy review, and feedback session | Project Manager, SteCo |
| User Feedback Collection | Form / Interview | During beta testing | Usability and feature suggestions | End Users, QA Lead |
| Final Presentation | In-person | End of semester | Full system demo and documentation | Team & Instructor |

This communication plan ensures that:

- All project members are informed of progress, deadlines, and issues.
- The instructor receives consistent progress updates.
- End users are engaged early in the development for better usability outcomes.
- Miscommunication and redundant work are minimized through centralized data and clear reporting structure.

# APPENDIX

- Interview Questions for Requirement Gathering
- Sample UI Mockups

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